

DAVENPORT, IA, USA

WMO: 744550

Lat: 41.614N

Lon: 90.591W

Elev: 229

StdP: 98.61

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 94982

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-18.2	-25.3	0.4	-21.3	-22.6	0.5	-18.0	13.5	-3.0	12.3	-5.7	3.8	300	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.7	23.8	31.3	23.4	30.0	22.6	25.6	30.8	24.7	29.7	23.8	28.5	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	29.0	23.0	18.2	28.1	22.2	17.3	27.1	79.7	30.7	75.9	29.9	72.3	28.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.7	9.2	-26.3	34.8	3.8	1.4	-29.0	35.8	-31.2	36.6	-33.3	37.4	-36.1	38.5		
(5)			-26.6	27.3	3.6	0.7	-29.2	27.9	-31.3	28.3	-33.3	28.7	-36.0	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-5.5	-4.1	3.8	10.3	17.0	22.0	23.3	22.1	18.7	11.4	4.1	-2.4		
	DBStd	11.58	7.07	7.07	6.58	5.43	5.02	3.22	3.25	2.85	4.54	5.66	5.81	6.51		
	HDD10.0	1784	481	397	212	62	6	0	0	0	1	50	191	384		
	HDD18.3	3508	739	628	451	246	87	6	3	3	50	226	428	641		
	CDD10.0	1831	0	1	22	71	222	359	411	375	262	94	13	1		
	CDD18.3	514	0	0	2	4	45	115	156	120	62	12	0	0		
	CDH23.3	4757	0	0	13	67	416	1054	1521	1015	564	108	0	0		
	CDH26.7	1526	0	0	1	8	117	344	553	297	181	24	0	0		
(14) Wind	WSAvg	4.4	5.1	5.1	5.2	5.5	4.8	3.9	3.1	2.9	3.4	4.2	4.9	4.8		
(15) Precipitation	PrecAvg	953	34	37	63	93	120	133	107	110	75	66	57	46		
	PrecMax	1315	65	96	142	177	251	355	287	244	222	175	155	96		
	PrecMin	488	9	3	13	34	3	26	10	28	20	12	7	9		
	PrecStd	190	17	26	42	38	56	79	65	57	47	40	38	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	17.3	25.2	27.9	32.6	33.0	35.1	33.0	32.7	29.6	21.3	15.3		
		MCWB	7.1	11.9	17.0	16.6	21.8	23.5	25.8	24.0	23.3	21.5	13.9	12.7		
	2%	DB	8.7	11.9	20.7	25.5	29.7	32.0	33.0	31.2	30.8	26.4	17.9	11.7		
		MCWB	6.2	8.4	14.3	16.2	20.3	23.4	24.7	23.8	22.1	18.6	12.7	8.8		
	5%	DB	5.6	8.1	17.2	22.5	27.6	30.3	31.2	29.8	28.8	23.6	15.4	8.7		
		MCWB	3.7	5.2	12.5	14.7	18.9	22.3	24.1	23.3	21.0	17.0	10.9	6.8		
	10%	DB	3.3	5.0	13.6	20.1	25.6	28.6	29.7	28.3	26.8	20.8	12.9	6.1		
		MCWB	1.7	3.0	9.4	12.9	18.0	21.7	23.2	22.1	19.8	15.4	9.4	4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	12.9	17.9	18.6	23.1	25.7	27.2	26.1	24.5	22.5	15.7	13.7		
		MCDB	10.7	15.3	24.1	24.4	30.2	31.3	32.0	30.9	30.6	28.2	18.6	15.4		
	2%	WB	6.0	8.7	15.5	17.2	21.6	24.7	25.9	25.1	23.5	20.5	13.5	9.1		
		MCDB	8.4	11.7	19.6	23.4	27.7	30.0	31.2	29.9	28.5	24.6	17.0	10.6		
	5%	WB	3.7	5.7	12.8	15.6	20.4	23.6	25.1	24.0	22.3	18.4	11.7	6.7		
		MCDB	5.6	8.0	16.4	21.4	25.6	28.7	30.2	28.3	26.8	21.8	15.2	8.6		
	10%	WB	1.8	2.9	9.8	13.9	19.2	22.5	24.1	23.0	21.2	16.0	9.4	4.0		
		MCDB	3.1	4.9	13.5	19.0	24.1	27.4	28.7	26.9	25.4	20.2	12.7	5.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	10.1	11.9	11.5	10.7	10.7	10.8	12.4	11.5	10.1	7.8		
		MCDBR	10.3	13.1	14.0	15.9	13.3	12.0	11.5	11.6	13.1	14.1	13.1	10.4		
	5% WB	MCWBR	8.4	9.7	9.0	8.9	6.5	5.9	5.4	5.9	6.0	7.6	8.8	8.3		
		MCWBR	9.8	12.2	12.4	13.4	11.7	10.6	10.2	9.9	11.0	11.1	12.2	10.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.292	0.302	0.342	0.386	0.419	0.423	0.441	0.421	0.392	0.344	0.302	0.290		
		taud	2.403	2.391	2.373	2.259	2.231	2.270	2.205	2.263	2.315	2.449	2.502	2.435		
	Ebn at Noon	Ebn at Noon	861	908	905	883	859	854	833	839	840	848	847	832		
		Edh at Noon	79	94	108	131	138	133	141	129	114	89	72	71		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.76	2.75	3.74	4.67	5.47	6.14	6.09	5.39	4.56	3.00	1.99	1.46		
		RadStd	0.21	0.28	0.32	0.40	0.40	0.54	0.40	0.35	0.38	0.35	0.17	0.15		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page